

Required Background

In addition to the concepts of Chapter I, this chapter requires some basic linear algebra concepts, briefly reviewed in the following.

► 15.1 Vectorised Matrices and the Kronecker Product

This chapter revolves around inferring matrix-valued objects. For this purpose, a matrix $A \in \mathbb{R}^{N \times M}$ will frequently be treated as a collection of NM real numbers arranged into a rectangular table, rather than an object associated with an algebraic structure. For a real matrix $A \in \mathbb{R}^{N \times M}$, the symbol $\vec{A} \in \mathbb{R}^{NM}$ will then denote the *vector* arising from stacking the elements of A row after row (“row-major” indexing¹). It would be possible to index the elements of this vector with a single number running from 1 to NM . However, to avoid confusing translations between index sets, the elements of this vector will be indexed by the same index set $(ij) \subset \mathbb{N} \times \mathbb{N}$ as the matrix itself.

On some rare occasions it will be necessary to use the inverse operation of vectorisation – to re-shape a vectorised matrix back into rectangular form. This will be denoted by the $\natural$ symbol:

$$A = \natural \vec{A}. \quad (15.1)$$

The *Kronecker product* $A \otimes B$ of two matrices $A \in \mathbb{R}^{N_A \times M_A}$, $B \in \mathbb{R}^{N_B \times M_B}$ is a matrix of size $N_A N_B \times M_A M_B$ with elements

$$[A \otimes B]_{ij,kl} = [A]_{ik}[B]_{jl}. \quad (15.2)$$

This matrix maps from $\mathbb{R}^{M_A \cdot M_B}$, the space of vectorised real $M_A \times M_B$ matrices, to $\mathbb{R}^{N_A \times N_B}$, the space of vectorised $N_A \times N_B$ matrices. For $\vec{C} \in \mathbb{R}^{M_A \cdot M_B}$, and $C = \natural \vec{C}$, we have

$$[(A \otimes B)\vec{C}]_{ij} = \sum_{k\ell} [A]_{ik}[C]_{k\ell}[B]_{j\ell} = [\overline{ACB^T}]_{ij}. \quad (15.3)$$

¹This is *not* that same as stacking the matrix column-by-column (“column-major”). The latter is used in Fortran and thus also by Matlab’s $A(:)$ operator. Numpy’s $A.ravel()$ allows the user to specify the order, but defaults to row-major indexing, which is also a standard used in C. Under column-major indexing, Eq. (15.3) changes to

$$(A \otimes B)\vec{C} = \overline{BCA^T},$$

which would cause all sorts of havoc down the line.

In this sense the Kronecker product “translates” between matrix multiplication and vectorisation. The Kronecker product has a number of analytically useful properties,² which mostly arise from the relation

$$(A \otimes B)(C \otimes D) = AC \otimes BD. \quad (15.4)$$

Among them are

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1},$$

$$|A \otimes B| = |A|^{\text{rk}(B)} \cdot |B|^{\text{rk}(A)}, \quad (15.5)$$

$$(A \otimes B) = (V_A \otimes V_B)(D_A \otimes D_B)(V_A \otimes V_B)^{-1}, \quad (15.6)$$

$$(A \otimes B)^{\top} = A^{\top} \otimes B^{\top}, \quad (15.7)$$

where $|A|$ is the matrix determinant of A , $\text{rk}(A)$ is the rank of A , and $A = V_A D_A V_A^{-1}$, $B = V_B D_B V_B^{-1}$ are the eigenvalue decompositions of A and B , respectively. These properties only hold assuming that these decompositions and inverses exist.

Exercise 15.1 (easy but instructive, solution on p. 362). *Prove Eqs. (15.4)–(15.7). Show that Eq. (15.6) is indeed the eigen-decomposition of $A \otimes B$.*

► 15.2 Positive Definite Matrices

A real matrix $A \in \mathbb{R}^{N \times N}$ will be called **SPD** if it is symmetric ($A = A^{\top}$) and positive (semi-) definite. It is *positive (semi-) definite* if $v^{\top} A v > 0$ (if $v^{\top} A v \geq 0$) for all non-zero vectors $v \in \mathbb{R}^N$. Like all symmetric real matrices, **SPD** matrices have orthogonal eigenvectors. Strictly positive definite matrices have strictly positive real eigenvalues, and are thus invertible.

► 15.3 Frobenius Matrix Norm

There are many norms³ on the space of real matrices $A \in \mathbb{R}^{N \times M}$, some with certain analytical advantages over others. The *Frobenius norm* is defined by

$$\|A\|_F^2 := \text{tr}(A^{\top} A) = \sum_{i=1}^N \sum_{j=1}^M A_{ij}^2 = \vec{A}^{\top} \vec{A} = \|\vec{A}\|_2^2,$$

where $\|\cdot\|_2$ is the standard Euclidean (ℓ^2) vector-norm. Of particular importance will be the generalised, *weighted Frobenius norm*, using two symmetric positive definite matrices V, W :

$$\|A\|_{F,V,W}^2 := \text{tr}(A^{\top} V^{-1} A W^{-1}) = \vec{A}^{\top} (V \otimes W)^{-1} \vec{A}. \quad (15.8)$$

³ A matrix norm $\|A\| \in \mathbb{R}$ has the property that

- ⊕ $\|A\| \geq 0$,
- ⊖ $\|A\| = 0$ iff $A = 0$,
- ⊕ $\|\alpha A\| = |\alpha| \|A\|$ for all $\alpha \in \mathbb{R}$,
- ⊕ $\|A + B\| \leq \|A\| + \|B\|$.

Exercise 15.2 (easy, solution on p. 362). *Using Eq. (15.6), show that the Kronecker product of two symmetric positive definite matrices is itself **SPD**, then use this result to show that Eq. (15.8) does indeed define a matrix norm.*

► 15.4 Singular Value Decomposition

Every real matrix⁴ $B \in \mathbb{R}^{M \times N}$ has a *singular value decomposition*

⁴ There is an analogous formulation for complex matrices, but the chapter only involves real matrices.

(SVD)

$$B = Q\Sigma U^T$$

with orthonormal⁵ matrices $Q \in \mathbb{R}^{N \times N}$, $U \in \mathbb{R}^{M \times M}$, whose columns are called the *left- and right- singular vectors*, respectively, and a rectangular diagonal matrix⁶ $\Sigma \in \mathbb{R}^{N \times M}$ which contains non-negative real numbers called *singular values* of B on the diagonal. Assume, w.l.o.g., that $N \geq M$ and the diagonal elements of Σ are sorted in descending order, and Σ_{rr} with $r \leq M$ is the last non-zero singular value. Then Q can be decomposed into its first r columns, Q_+ , and the (potentially empty) $N - r$ columns, Q_- as $Q = [Q_+, Q_-]$, and similarly $U = [U_+, U_-]$ for the columns of U . The SVD is a powerful tool of matrix analysis:

⁵ That is, $Q^T Q = I_N$ and $U^T U = I_M$.

⁶ That is, $\Sigma_{ij} = 0$ if $i \neq j$.

- ⊖ r equals the *rank* of B : $\text{rk}(B) = r$;
- ⊖ the columns of U_- span the *null space* (the *kernel*) of B (those of U_+ the *pre-image* of B);
- ⊖ the columns of Q_+ span the *range* (the *image*) of B (those of Q_- the *co-kernel* of B);
- ⊖ the matrix $\tilde{B} = \arg \min_{\text{rk}(\tilde{B})=k \leq r} \|B - \tilde{B}\|_F$ (“the best rank- k approximation to B in Frobenius norm”) is given by $\tilde{B} = Q\tilde{\Sigma}U^T$, where $\tilde{\Sigma}$ equals Σ except that $\tilde{\Sigma}_{ii} = 0$ for $i > k$;
- ⊖ the matrix $V = \arg \min_{V^T V = I} \|B - V\|_F$ (“the orthonormal matrix closest to B in Frobenius norm”) is $V = QU^T$.

► 15.5 Matrix Identities

For matrices A, U, C, V , if all the inverses in the following equation exist, then the *matrix inversion lemma*⁷ holds:

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}. \quad (15.9)$$

It is helpful if U is a “skinny” matrix (i.e. $U \in \mathbb{R}^{N \times M}$, $M \ll N$), and the inverse of A is known, because then the inverse of the larger $N \times N$ left-hand side of Eq. (15.9) can be found from the smaller $M \times M$ inverse on the right-hand side.

⁷ The matrix inversion lemma has been ascribed to many different authors, including M. Woodbury, J. Sherman and W. Morrison, M. Bartlett, and I. Schur. It is likely that the result is even older than any of the corresponding works. Some historical background, along with interesting additional numerical analysis, can be found in a review by Hager (1989).

The inverse of the block matrix

$$A := \begin{bmatrix} P & Q \\ R & S \end{bmatrix}, \text{ if it exists, is } A^{-1} = \begin{bmatrix} \tilde{P} & \tilde{Q} \\ \tilde{R} & \tilde{M} \end{bmatrix}, \quad (15.10)$$

with

$$\begin{aligned} M &:= (S - RP^{-1}Q)^{-1}, & \tilde{P} &:= P^{-1} + P^{-1}QMRP^{-1}, \\ \tilde{Q} &:= -P^{-1}QM, & \tilde{R} &:= -MRP^{-1}. \end{aligned}$$

This relationship is often attributed to I. Schur (1917). The term P is sometimes called the *pivot block*, the term M , and others of its general form, are widely called *Schur complements*.⁸ This is due to its use in the Schur determinant formula, also simply known as the *determinant lemma*, which follows from the above:

$$|A| = |P| \cdot |S - RP^{-1}Q|.$$

It can also be re-phrased in the notation of the matrix inversion lemma above:

$$|A + UCV| = |A| \cdot |C| \cdot |C^{-1} + VA^{-1}U|. \quad (15.11)$$

The related result for block matrices is⁹

$$\begin{aligned} A \text{ non-singular} &\Rightarrow \det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(A) \det(D - CA^{-1}B), \\ D \text{ non-singular} &\Rightarrow \det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(D) \det(A - BD^{-1}C). \end{aligned}$$

⁸ An introduction to Schur complements, including motivations for the terminology and a discussion of applications and related concepts, is provided by Cottle (1974).

⁹ Lütkepohl (1996), §4.2.2, Eq. (6)